Dept. of Computer Science and Engineering

Jashore University of Science and Technology

Faculty of Engineering and Technology - FET

Lab Report : 02

ER Diagram and Database structure

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<u>Course Details:</u>	**<u>Student Details:</u>**
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Project Name: WellBeing – A Mental Health Improver Platform

Project Description:

WellBeing is an interactive web platform crafted to empower individuals on their mental health* Comprehensive Mood Tracking: Users log daily moods via an intuitive emoji slider or dropdown (ranging from Very Happy to Very Sad), optionally adding contextual notes or journal reflections.

* Insightful Analytics Dashboard: Visualize mood trends over configurable timeframes using interactive charts, view weekly average scores, and monitor consistency streaks to foster self - awareness.

* Tailored WellB eing Tips: Leverage a curated library of self ? care suggestions—daily affirmations, breathing exercises, and mindfulness prompts—dynamically recommended based on recent emotional data.

* Private Journaling Module: A secure diary feature where users can record thoughts,tag entries, andsearch or filter past reflections for personal review.

* Therapeutic Resource Hub : Browse expertprofiles complete with specialties, availability, and contact options. Access emergency hotlines andcommunity support links.

* Appointment Scheduling: Seamless booking interface to request consultations, with status tracking—pending, confirmed, or cancelled—and calendar integration.

Under the hood, a relational database ensures data integrity, while session management and encryptionuphold user privacy. Responsive layouts guarantee accessibility across devices, with emphasis on calmingUI and user-centric navigation.

Functional Requirements:

Secure Authentication: Signup and login flows with encrypted credentials, session handling, and validation.

Responsive Home Interface: Clean landing page with guided CTAs and adaptive navigation menu.

Mood Entry Workflow: Interactive form for mood submission, storing timestamped entries in the database.

Analytics Dashboard: Endpoint-driven chart rendering for mood history, averages, and streak indicators.

Tips & Activities Section: Dynamic suggestion listing and optional activity logging for self-care tasks.

Journal Management: Create and retrieve rich-text diary entries, with search and preview functionalities.

Resources Directory: Display profiles for mental health professionals and vital support links.

Booking Module: Submit and manage appointment requests with relational linking between users and professionals.

Database Schema Details:

This database schema represents a Mental Health Management System, designed to support user journaling, mood tracking, therapy appointments, and personalized suggestions. Here's a concise summary:

Users Table: Stores personal user details including name, email, age, gender, and account creation date.

Journals Table: Allows users to write journal entries with associated timestamps for emotional tracking.

Mood Entries Table: Users log their moods and mood scores along with optional notes and timestamps.

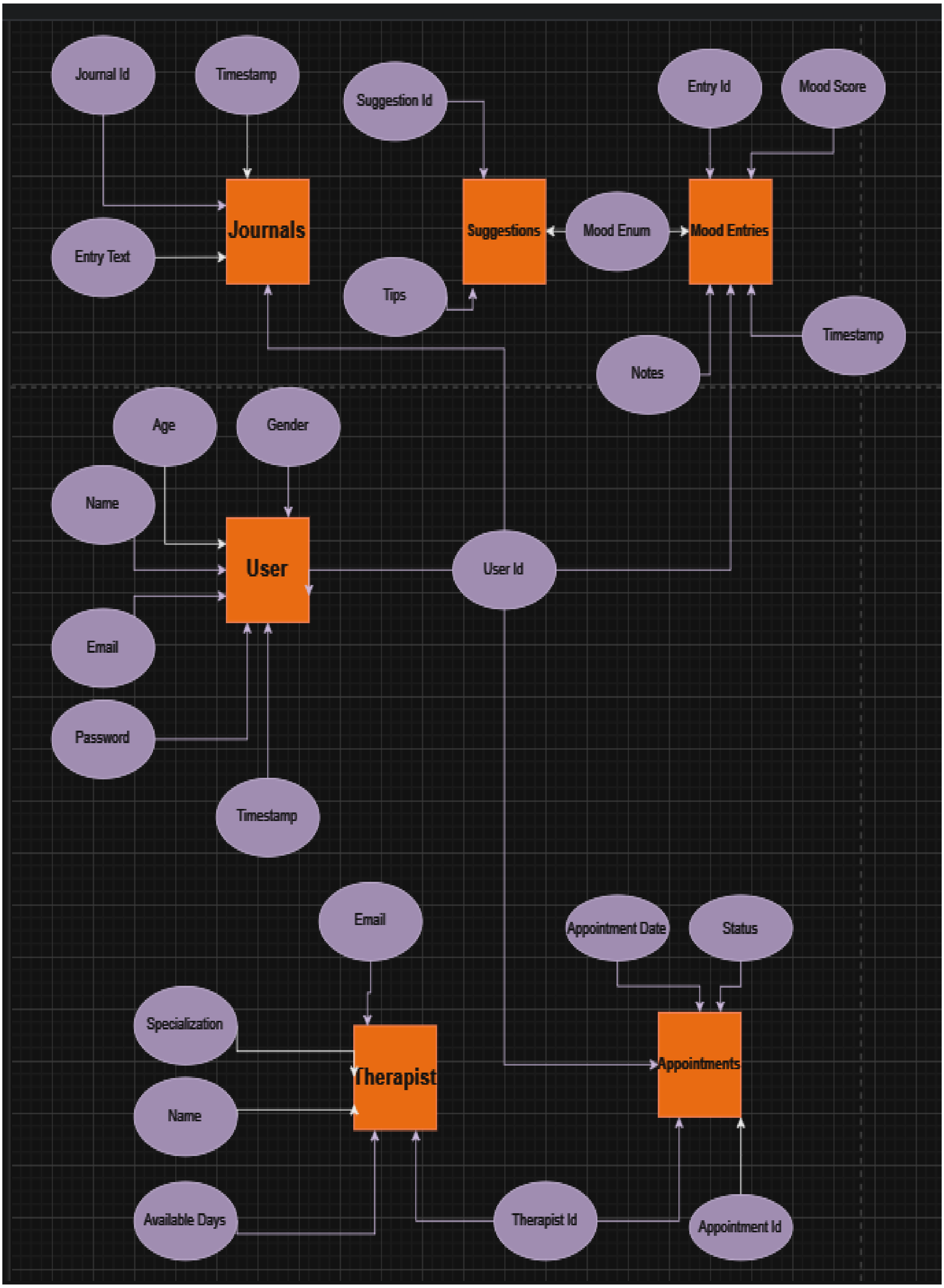
Suggestions Table: Provides mental health tips or advice based on mood entries.

Therapists Table: Contains therapist details such as name, specialization, email, and availability.

Appointments Table: Connects users with therapists, tracking appointment dates and statuses.

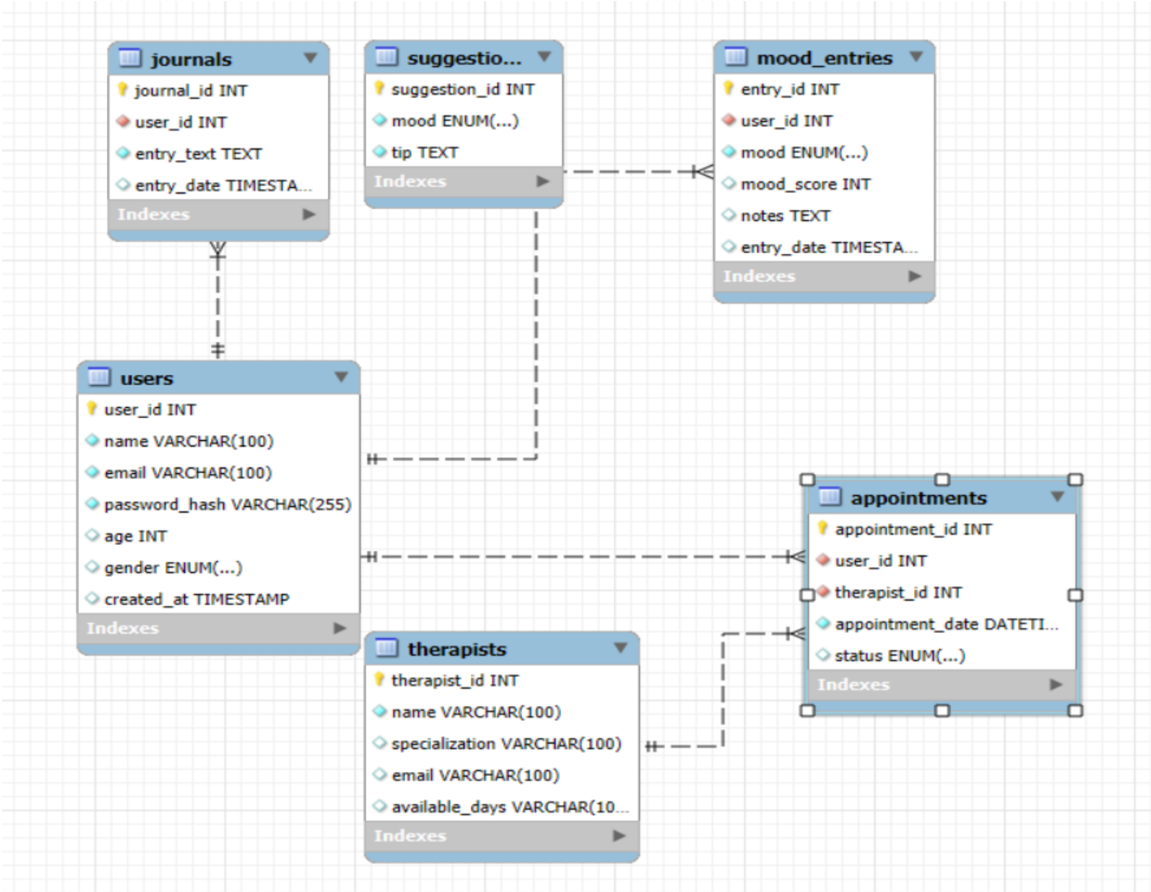
Relationships:
Users can have multiple journal entries, mood logs, and appointments.
Each appointment links a user to a therapist.
Suggestions may be generated in response to mood entries.
Enums: Gender, mood types, and appointment statuses are standardized using ENUM fields.
Data Types: Includes VARCHAR for text, INT for identifiers and scores, TEXT for longer inputs, and timestamps for time-based tracking.
Use Case: Ideal for supporting mental well-being by combining self-reflection, emotional monitoring, professional therapy, and tailored advice.

Database Design:
Figure 1. EER Diagram



Database Design (MySQL)

Figure 2. EER Diagram



SQL Schema Sample

```sql
CREATE DATABASE IF NOT EXISTS wellbeingjust;
USE wellbeingjust;

-- 1. Users table
CREATE TABLE users (
  user_id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100) NOT NULL,
  email VARCHAR(100) UNIQUE NOT NULL,
  password_hash VARCHAR(255) NOT NULL,
  age INT,
  gender ENUM('Male','Female','Other') DEFAULT 'Other',
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 2. Mood Entries
CREATE TABLE mood_entries (
  entry_id INT AUTO_INCREMENT PRIMARY KEY,
  user_id INT NOT NULL,
  mood ENUM('Very Happy','Happy','Neutral','Sad','Very Sad') NOT NULL,
  mood_score INT CHECK (mood_score BETWEEN 1 AND 5),
  notes TEXT,
  entry_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CASCADE
);

-- 3. Journals
CREATE TABLE journals (
  journal_id INT AUTO_INCREMENT PRIMARY KEY,
  user_id INT NOT NULL,
  entry_text TEXT NOT NULL,
  entry_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CASCADE
);
```

```sql
-- 4. Suggestions (Tips)
CREATE TABLE suggestions (
  suggestion_id INT AUTO_INCREMENT PRIMARY KEY,
  mood ENUM('Very Happy','Happy','Neutral','Sad','Very Sad') NOT NULL,
  tip TEXT NOT NULL
);

-- 5. Therapists
CREATE TABLE therapists (
  therapist_id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100) NOT NULL,
  specialization VARCHAR(100),
  email VARCHAR(100),
  available_days VARCHAR(100)
);

-- 6. Appointments
CREATE TABLE appointments (
  appointment_id INT AUTO_INCREMENT PRIMARY KEY,
  user_id INT NOT NULL,
  therapist_id INT NOT NULL,
  appointment_date DATETIME NOT NULL,
  status ENUM('Pending','Confirmed','Cancelled') DEFAULT 'Pending',
  FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CASCADE,
  FOREIGN KEY (therapist_id) REFERENCES therapists(therapist_id) ON DELETE CASCADE
);
```